Spectrum saturation, problematic WiFi to cellular handover, high latency, and chipset cost of 4G technologies are some of the roadblocks for a cluster of emerging technologies. 5G promises high data rates, low latencies, reliable connections and supports 1 million devices per square kilometer, opening up a number of new applications.

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MEDICINE

Remote Surgery: Low latency and high reliability means expert surgeons can save lives by performing intricate surgeries remotely, using robotics.

Continuous Monitoring: Low cost wearables with reduced energy requirements can monitor patients in real time, for telltale signs of any problems.

Smart Medicines: Pharmaceuticals can inform the doctor if the patient skips a dose, and sensors in bandages can monitor how well wounds are healing.

Asset Management: Hospitals can keep track of medicines, facilities and personnel better, routing them to where they are most needed.

Biomedical Imaging: Low cost 5G will allow for high speed transfer and sharing of biomedical images, including 3D scans, to multiple parties in realtime.

RETAIL

Biometrics in Shopping: Facial recognition or other biometric identification will allow users to checkout purchases through cashless transactions.

Stock management: Embedded sensors can notify personnel or other systems when stocks of a particular item are running low.

Quicker purchases: Chatbots, AI assistants and improved connectivity mean a better and faster experience for the end user.

Efficient space usage: Tracking movement of goods combined with data analytics can allow for better space management in stores.

Smart Advertisements: Billboards, hoardings and banners can react to individual users, or change according to movement of the goods.

LOGISTICS

Realtime Tracking: Continuous, real time tracking of parcels from the factory floor to the consumer will be possible for the first time.

Foodstuff quality: Buyers can be assured of the origin and freshness of foodstuff, from fruits to fish, because of continuous tracking.

Optimised supply chains: Movement of goods can be optimised based on feedback from thousands of sensors across in various stages of the supply chain.

Minimise losses: Cheap sensors can also identify exactly when and where a parcel was damaged, helping pin responsibility, and reduce losses.

Better routing: Embedded sensors can provide truckers or platoons with information on weight, fuel efficiency and temperatures, allowing for better routing.

SMART CITIES

Utility Metering: Water, gas and electricity can be metered in real time, at levels of accuracy not possible before, at multiple stages of the distribution.

Flexible Distribution: Large scale smart power grids can even out variability, and make green power sources such as wind and solar energy viable.

Demand Response: Digital sensors will allow power distribution companies to respond quicker to demand, scaling generation as required.

Domotics: IoT will spread to more appliances and objects within the household, allowing for smarter homes within smart cities.

Smart Lighting: Instead of street lamps staying on throughout the night, they can increase to full power only when there is vehicle movement.